

Muhammad Abdullah

Senior Software Engineer

📞 92-330-9519366 ✉️ malik3661@live.com 📍 Islamabad, Pakistan
📄 muhammad-abdullah3661 🔗 Genesis3661



PROFESSIONAL SUMMARY

Senior AI Data Engineer & Technical Lead (5+ yrs) building enterprise data pipelines, warehouse architectures, and production AI systems. Specializes in end-to-end ETL, conversational AI, and agentic developer tools.

SKILLS

- Languages & Backend: TypeScript JavaScript Python Java SQL NestJS Node.js REST APIs Angular
- Data & Warehousing: AWS Redshift PostgreSQL ETL & Streaming Pipelines Schema Design Data Validation
- AI & ML: LangChain LangGraph DeepSeek Pinecone RAG Agentic Systems
- Cloud & DevOps: AWS Docker Linux CI/CD
- System Design: Concurrency & Backpressure Queue Architecture Performance Optimization
- Mobile (Android): Android SDK Kotlin Java Jetpack Room MVVM Maps & Geolocation REST Integration

EXPERIENCE

01/2024 - Current
Islamabad, Pakistan

- Senior Software Engineer
Auxiliotech
 - Architected and delivered MailFlow, a production-grade data ingestion platform processing distributor Excel datasets (Stock, Sales, Transit, Actuals, IMS, TMS Estimates) with automated validation, transformation, and exception resolution before publishing clean data to AWS Redshift – **reducing the monthly reporting cycle from weeks of manual work to minutes while eliminating human error.**
 - Engineered a memory-bounded, multi-stage streaming pipeline (extraction → structuring → validation → exception checking → insertion) using producer-consumer workers with bounded queues and backpressure, **sustaining ingestion of millions of rows at flat memory usage regardless of file size.**
 - Maximized AWS Redshift insert throughput with a custom concurrency limiter (semaphore) and adaptive, query-size-aware batching that estimates SQL statement size against Redshift's 16 MB limit and dynamically splits work into parallel batches only when required.
 - Designed and deployed a conversational AI assistant that lets business stakeholders query company databases in natural language, **eliminating manual dashboard and report requests and cutting decision-making turnaround from days to minutes.**
 - Built an AI-powered developer coding assistant on an agentic architecture of specialized, collaborating agents aligned to internal codebases, databases, and engineering standards – **improving development productivity by ~40%** and reducing boilerplate/base-template generation from hours to minutes.
 - Used SQLite in WAL mode as a tuned out-of-core store (custom PRAGMAs, manual checkpointing) to offload unbounded intermediate state from RAM, backed by a persistent file queue with controlled concurrency, cancellation, failure recovery, and deduplication that survives server restarts without data loss.
 - Cut external validation lookups by 60%** through SHA-256 deduplication of repeated values, maintaining a value → row mapping with a "checked" cache that eliminates redundant API/database calls during exception resolution.
 - Implemented schema-flexible Redshift staging using SUPER columns with JSON_PARSE, plus schema-aware validation against the live database schema (type, precision/scale, nullability) for accurate, constraint-driven data-quality checks.
 - Designed, optimized, and maintained large-scale PostgreSQL and Redshift schemas for

EXPERIENCE

enterprise clients – resolving problematic foreign-key dependencies and restructuring tables to accelerate analytics and AI query performance.

- Designed and maintained scalable, cloud-native backend services (NestJS, Node.js, TypeScript, Docker, REST) deployed across Linux staging and production, converting front-end-driven workflows into high-performance, multi-threaded services for greater throughput and operational scalability.
- Delivered multiple workflow-automation applications for large multinational pharmaceutical clients – covering agreement management, approval chains, dynamic calculations, and automated status notifications – replacing manual processes with secure, auditable digital workflows that significantly reduced turnaround time and human error.
- Led stakeholder meetings to translate business requirements into data-architecture designs, technical roadmaps, and delivery milestones, and contributed to architecture reviews and sprint estimation to improve delivery predictability.
- Established code-review standards, QA checklists, and engineering best practices, and mentored engineers across data engineering, backend, and AI integration – measurably improving software quality, release confidence, and team output.

09/2018 – 07/2021
Islamabad, Pakistan

• Mobile Application Developer Resatech

- Designed, developed, and deployed an Android application for 5G network quality testing in live field environments, owning the full product lifecycle from requirement gathering to production release.
- Integrated Google Maps APIs for real-time device location tracking and routing, enabling precise geospatial monitoring throughout active network test sessions.
- Automated device-level test actions including call initiation, SMS dispatch, upload/download tests, and performance logging while devices roamed across the city – eliminating manual test execution entirely.
- Built background services to continuously capture network performance metrics and securely transmit telemetry to a central server for automated reporting and analysis.
- Optimized the application for continuous GPS tracking, network activity logging, and background data sync while maintaining minimal battery consumption – critical for long field test sessions.
- Collaborated with backend teams to design secure REST API endpoints for efficient, reliable data transfer between field devices and the reporting infrastructure.
- Integrated push notifications to coordinate field testers in real time, reducing miscommunication and improving operational efficiency during city-wide test runs.
- Delivered a mission-critical tool that streamlined 5G testing operations and accelerated network quality analysis for telecoms infrastructure decisions.

EDUCATION

06/2014 – 12/2018
Computer Science

• Bachelor of Computer Science COMSATS

PROJECTS

Distribution Data Reporting

Distribution Data Reporting (DDR) Platform An event-driven platform automating the full distributor reporting chain. Files emailed by global distributors land on a shared EFS volume, where a watcher enqueues them into a persistent queue for automated ingestion pipeline, validation, error/exception checking, approval routing, and publication to AWS Redshift – escalating to users only on unresolved exceptions. **Collapses a weeks-long, multi-team manual process into a hands-off pipeline that moves data from inbox to warehouse in minutes.**

Stack: NestJS, Node.js, TypeScript, Angular, PostgreSQL, AWS Redshift, SQLite, Chokidar, AWS EFS, Docker

Virtual Business Analyst

A natural-language interface that lets business stakeholders query company databases directly, without waiting on analysts or manual dashboards. Turns ad-hoc data requests that previously took days into instant answers, putting real-time decision-making in the hands of non-technical users. Stack: NestJS, Node.js, TypeScript, LangChain, DeepSeek, Pinecone, PostgreSQL

- A natural-language interface that lets business stakeholders query company databases directly, without waiting on analysts or manual dashboards. Turns ad-hoc data requests that previously took days into instant answers, putting real-time decision-making in the hands of non-technical users.
- **Stack: NestJS, Node.js, TypeScript, LangChain, DeepSeek, Pinecone, PostgreSQL**

Rapid Coder

An agentic coding assistant built from specialized, collaborating agents aligned to internal codebases, databases, and engineering standards. Accelerates development by generating standards-compliant code and base templates in minutes, improving team productivity and consistency, with continuous improvement through live updates.

Stack: NestJS, Node.js, TypeScript, LangGraph, DeepSeek, Pinecone

Promotional Expense Management

- A workflow-driven platform for large pharmaceutical clients to manage annual promotional budgets end to end – defining yearly budgets, raising and approving expense requests against them throughout the year, and formally closing requests on completion. Fully dynamic Angular forms, approval chains, and calculations replaced a manual, error-prone process with an auditable, real-time view of budget allocation and spend.
- **Stack: JavaScript, Node.js, PostgreSQL, Angular**

Nylytics

An Android-based drive-test tool for measuring 5G network performance in the field. Automatically runs call, SMS, and download/throughput tests along predefined drive routes – built on the Maps APIs so field drivers simply follow the on-screen path – then uploads results to an admin dashboard for real-time network-quality analysis. Replaced manual, hardware-heavy drive testing with an accessible, automated mobile solution.

Stack: Android SDK, Java, Google Maps SDK, Directions API, Places API, REST APIs